Week 38: Seams

Adult guide built from the problem files · Unpiloted. The guide's physical pretest is unperformed.

This week's adult guide is week-38-facilitator.pdf, whose source is in editable/facilitator-src/. These problem files hold each problem's one-sentence idea, not its answers.

Problems and routes

Problem	K–1	2–3	4–5	route
Trace every edge of A and B	P1	P1	P1	first
Two dots one edge trip can or cannot reach	P2	P2	P2	first
Four dots to match each ring picture	P3	P3	P3	first
Predict, then cut along the middle line	P4	P4	P4	first
Explain the cut with U and L	P5	P5	P5	return
Which ring pictures after the cuts?	P6	P6		return
Cut a new B one third of the way across	P7	P7		return
Carry an arrow once around each band			P6	return
Ring pictures after the cuts, pieces versus edges			P7	return

First route. K–1 P1–P4; Grades 2–3 P1–P4; Grades 4–5 P1–P4. **Return visits.** K–1 P5–P7; Grades 2–3 P5–P7; Grades 4–5 P5–P7.

Key, one entry per problem

Trace every edge of A and B

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. A strip joined matching (A) has two boundary circles; joined with a half-turn (B) it has one.

Two dots one edge trip can or cannot reach

K–1 P2 · Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

Idea. Two edge dots share a trip exactly when they lie on the same boundary circle; on B, with one circle, every pair does.

Four dots to match each ring picture

K–1 P3 · Grades 2–3 P3 · Grades 4–5 P3 · first route

Idea. With b boundary circles, four dots can be grouped exactly as the partitions of 4 into at most b parts.

Predict, then cut along the middle line

K–1 P4 · Grades 2–3 P4 · Grades 4–5 P4 · first route · wording differs by band

Idea. Cutting A along its middle makes two annuli; cutting B makes one piece, an annulus with two edges.

Explain the cut with U and L

K–1 P5 · Grades 2–3 P5 · Grades 4–5 P5 · return route · wording differs by band

Idea. The seam sends the halves U and L to themselves on A (two pieces) and swaps them on B (one piece).

Which ring pictures after the cuts?

K–1 P6 · Grades 2–3 P6 · return route

Idea. After the cuts A gives four boundary circles and B two, so A's pieces make every picture and B's only those with at most two rings.

Cut a new B one third of the way across

K–1 P7 · Grades 2–3 P7 · return route

Idea. Cutting B one third of the way across goes round twice and leaves a thin Moebius band linked with a longer two-edge band: three closed edges, so every picture but 1+1+1+1.

Carry an arrow once around each band

Grades 4–5 P6 · return route

Idea. An arrow slid once around A returns as it started; around B it returns reversed, because the seam joins with a half-turn.

Ring pictures after the cuts, pieces versus edges

Grades 4–5 P7 · return route

Idea. Which pictures are possible depends on the number of boundary circles, not the number of connected pieces.